

IMO(24th) – 1983

First Day

1. Find all functions f defined on the positive real numbers and taking positive real values that satisfies the following conditions:
 - (a) $f(xf(y)) = yf(x)$ for all positive real x, y ;
 - (b) $f(x) \rightarrow 0$ as $x \rightarrow +\infty$.
2. Let K be one of the two intersection points of the circles W_1 and W_2 . Let O_1 and O_2 be centres of W_1 and W_2 . The two common tangents to the circles meet W_1 and W_2 respectively P_1 and P_2 , the first tangent, and Q_1 and Q_2 the second tangent. Let M_1 and M_2 be the points of P_1Q_1 and P_2Q_2 respectively. Prove that $\angle O_1KO_2 = \angle M_1KM_2$.
3. Let a, b, c be positive integers satisfying $(a, b) = (b, c) = (c, a) = 1$. Show that $2abc - ab - bc - ca$ is the largest integer not representable as $xbc + yca + zab$ with non negative integers x, y, z .

Second Day

4. Let ABC be an equilateral triangle. Let E be the set of all points from segments AB , BC and CA (including A , B and C). Is it true that for any partition of the set E into two disjoint subsets, there exists a right angled triangle all of whose vertices belong to the same subset in the partition?
5. Prove or disprove the following statement: In the set $\{1, 2, 3, \dots, 10^5\}$ a subset of 1983 elements can be found that does not contain any three consecutive terms of an arithmetic progression.
6. If a, b, c are sides of a triangle, prove that $a^2b(a-b) + b^2c(b-c) + c^2a(c-a) \geq 0$ and determine when there is equality.

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